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Dear Editors of the *Journal of Computational Physics*,

We submit for your consideration the manuscript *"Reconstruction Order, Not Solver Family, Governs Control Accuracy: A Comparison of Neural and Classical HJB Solvers Against an Exact Reference"*.
The paper measures something that is usually asserted. Neural solvers for Hamilton–Jacobi–Bellman equations are proposed on the strength of the curse of dimensionality, and the comparison literature scores them against other solvers — which makes the reference exact by construction and its reported error zero. We score four families against a closed-form Riccati reference on a multi-asset optimal-execution HJB, and we report the *control* as well as the value. Three results follow, and each is stated as a rate rather than an observation:

- **Two propositions.** A degree-k reconstruction delivers gradient order k, but a symmetric-difference probe at step ε adds an independent $O(\varepsilon^2)$, so a probe that shrinks only like h caps a cubic reconstruction at order two. And for this problem the objective lost is *exactly* the R-weighted squared control deviation accumulated along the path taken — an identity, not the bound the envelope theorem would give. Together they explain why the control error is 3.2 to 26 times the value error, and they reduce to a prescription: raise the reconstruction order, and scale the probe with it.

- **A boundary, measured rather than assumed.** Where a grid scheme runs it is faster and more accurate; we also run a compiled kernel of the same scheme, which changes the comparison at the top dimension, and we run the classical mesh-free family the earlier version of this work had only cited — where it turns out to be *more* accurate than the neural solver at every dimension tested. A conclusion changed as a result, and the paper says so.

- **What it is worth.** On the paper's own real-market calibration, raising the reconstruction from multilinear to cubic is worth 2.7 to 8.4 basis points of notional across nine instruments, with the largest effect on the most liquid one.

The paper is a comparison and a measurement, and we have tried to make it useful in the way your readership would want: every table is generated from committed result files, the verification suite runs on a clean checkout, the scheme's own solver is the one used for the compiled timing so that implementation and algorithm are separated, and the limitations are stated where they bind rather than collected in a closing paragraph. Three of them are worth naming here. The grid sweep stops at $p = 5$; the neural results use one architecture family at stated budgets; and the non-polynomial benchmark we construct to answer the "your solution is quadratic" objection uses a manufactured source, which produces an exact solution but artificial economics.
We confirm that this manuscript is original, is not under consideration elsewhere, and that all authors have approved the submission. Declarations of competing interest, the CRediT statement and the data availability statement accompany the submission. Reproducibility is the point of the paper, so we have made every number traceable to a released file, and the repository is archived.

We would be glad to suggest referees if that is helpful, and we thank you for considering the work.

Sincerely,
Jingwen Zhang and Fei Ma